

SOC252H Intermediate Quantitative Methods
Winter 2025
Data Analysis Project

For this assignment, you will apply some of the methods we have learned so far in the course to conduct a statistical analysis with a real dataset. The dataset for this project is derived from the January 2024 wave of the Labour Force Survey (LFS). It contains a **random sample of Canadian residents aged 15-64 who were employed and were at work in the week prior to the survey**.

The dataset contains information on each respondent's basic demographics (e.g., sex, educational attainment, immigrant status), their labour force behaviour (e.g., how much they are paid, whether they have one or multiple jobs, their occupation, how many hours they work per week), and some information about their marriage and family.

Your goals for the data analysis project are to 1) come up with a research question that you can answer using these data, 2) conduct regression analysis to answer your question, and 3) write about your findings (including speculate about potential biases in your estimates). The final product will be a 1500-word research paper with an introduction and background, data and methods, results, and conclusion. Just like in academic articles we have read, you can include tables and graphs to help with the presentation of your findings.

Please use Times New Roman 12-pt font with double spacing throughout the essay. You will use APA style citations when you cite literature, and you should also include an APA-style reference list at the end of the essay. The word count does not include tables, graphs, or the reference list.

The research paper will be evaluated on five aspects: significance of the question, sufficiency of explanation of methods, rigor of data analysis, quality of conclusions, and general writing quality. Below are some examples of questions the grader may consider in evaluating each of the aspects.

1. Significance of the question.

- Is the question interesting and important? For example, if the author successfully finds the answer to their question, are there any practical problems it might solve? Could a policy maker be interested in the answer to the question – if so, how? Or is there a gap in the social science literature that this question speaks to?
- Is the question specific enough? Did the author clearly state what relationship they are interested in exploring?

2. Sufficiency of explanation of methods.

- Did the author clearly define their study population? As stated above, the dataset contains a random sample of Canadian residents aged 15-64 who were employed and were at work

in the week prior to the survey. However, the author may decide to further narrow down the population.

- Did the author explain what variables they are using for the study? Did they offer sufficient description of the variables so that readers have all the information necessary to interpret the findings? Did the author explain their rationale for using these variables?
- Did the author explain what regression they are using?

3. Rigor of the data analysis.

- Did the author use the appropriate regression for their research question and for the data?
- Are there clear flaws in the research design? For example, is there a clear violation of regression assumptions?
- Did the author interpret their findings correctly and sufficiently? Does it come through that they have fully mastered the regression technique and can confidently discuss different numbers from the regression output?
- Through their writing, did the author show thoughtfulness in their analysis and use rigorous language in their interpretation of findings?

4. Quality of conclusions.

- Did the author reach conclusions that clearly match the findings from their data analysis? Are the conclusions clearly relevant to the research question and highlight the significance of it?

5. General writing quality, citations, and formatting.

- Is the research paper easy to read and well organized?
- Did the author appropriately cite literature and include a complete reference list at the end of the paper?